

Particle Physics — Fermi's Golden Rule and the Dirac Equation

IMAPP lecture notes

Spring 2026

1 Fermi's golden rule

For a perturbation H_{int} and an initial state $|i\rangle$, the first-order transition rate into final states with density $\rho(E_f)$ is

$$\Gamma_{i \rightarrow f} = 2\pi |T_{fi}|^2 \rho(E_f), \quad T_{fi} = \langle f | H_{\text{int}} | i \rangle.$$

For relativistic particle physics, one uses the invariant matrix element $\mathcal{M}$ and the invariant phase-space measure instead of the frame-dependent T_{fi} .

2 The Dirac equation

The Klein-Gordon equation describes scalar particles but does not provide a positive-definite probability density for a one-particle interpretation. Dirac sought an equation first order in time and space:

$$i\partial_t \psi = (\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m) \psi.$$

Consistency with the relativistic dispersion relation requires

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij}, \quad \{\alpha_i, \beta\} = 0, \quad \beta^2 = 1.$$

These matrices are represented by 4×4 Dirac matrices, so ψ has four components.

Introducing $\gamma^0 = \beta$, $\gamma^i = \beta\alpha_i$, the covariant equation is

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

3 Particle and antiparticle solutions

The equation has positive- and negative-energy solutions. In the Feynman-Stueckelberg interpretation, negative-energy solutions are reinterpreted as positive-energy antiparticles propagating forward in time. The free spin-half solutions are written as spinors $u_s(p)$ for particles and $v_s(p)$ for antiparticles, with completeness relations of the form

$$\sum_s u_s(p) \bar{u}_s(p) = \not{p} + m, \quad \sum_s v_s(p) \bar{v}_s(p) = \not{p} - m.$$

Summary

The golden rule connects perturbative amplitudes to transition rates. The Dirac equation is Lorentz covariant, describes spin-1/2 fermions, and naturally predicts antiparticles.